

Highway 1 Soil Erosion Risk Analysis



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June 10, 2026

1. Introduction:

The Central Coast section of Highway 1 has a long, turbulent history of closures due to erosion. Its topography dominated by long steep slopes and prolonged sections with minimal plant cover, leave it susceptible to extremely high rates of erosion. It is also a coastal highway meaning that sea level rise causes sections of cliff and sometimes road to fall into the ocean as erosion encroaches from both above and below the highway. For the past 3 years from January of 2023 to January of 2026 the entirety of the Big Sur portion was closed due to Paul's Slide, a massive erosion event moving an estimated 500,000 cubic yards of soil (Lanham, 2024).

The fully closed section of road spanned 6.8 miles just north of San Simeon, a once thriving tourist town. Currently the town only has about 217 residents compared to the estimated 450 before Paul's Slide (Census Reporter), this steep drop in population is in direct relation to the closure of Highway 1. San Simeon relies almost entirely on tourism to prop up their economy so when the main reason visit is as a rest stop along Highway 1, a three-year long road closure means that countless businesses shut down. Local business owners feel they are being completely choked out by the closure and even now when the road is reopened, many businesses were too far gone to recover.

This is part of the reason it is so important to understand what areas are going to be at the highest risk for erosion and therefore closure. Erosion mitigation is almost entirely preventative as once the damage is done it is done, to effectively prevent erosion one must know where it is most likely to occur. Had more erosion measures been put into place, Paul's slide could have been prevented or at the very least mitigated against so a record-breaking three-year closure would have been far less likely.

Throughout the course of this report I will discuss how erosion risk was calculated for the research area, Santa Cruz to Cambria, using established methods, particularly the Revised Universal Soil Loss Equation (RUSLE). RUSLE is a well-documented and widely used soil loss equation with frequent uses in GIS software. recent applications can be seen

in the Abdelsamie et al. 2022 paper using it for risk assessment in Arid climates or in the 2018 paper by Fernandez using RUSLE in GIS to calculate soil erosion risk before and after the 2010 Pozo fire.

2. Methods:

2.1 Area of Study:

The study area encompassed a 252-mile stretch of Highway 1 from 35.1N to 37.1N, with a 2km buffer on either side of the road to analyze for erosion above and below the highway. All street data was collected from OpenStreetMap (OSM) via the Overpass API. This section of road is home to Monterey, Big Sur, and Santa Lucia which are all at very high risk of erosion due to their steep slopes and wet winters making it an ideal area of study. A buffer was defined to the above specifications in QGIS. A map of the area of study is included on page 4 in Figure 1.

2.2 RUSLE:

RUSLE is an equation that estimates mean annual soil loss based on five factors. The full equation is as follows: $A = R * K * LS * C * P$, with A being the output of mean annual soil loss. R is the variable for rainfall erosivity, this is measuring how much energy rain has to erode and transport soil particles and it is measured in Megajoules millimeters per hectare per hour per year. K is the soil erodibility factor which measures a soils ability to retain its soil particles, a higher K factor means a more erodible soil. LS is the length and steepness of the slope multiplied together, this is where topography is implemented into the equation. C is simply cover, this defines the level of plant cover of an area, a high C factor means less cover where a lower one means more cover. The P factor is management practice which has been omitted from this analysis. This is because spatial data on what erosion management has been done in the area is sparse and there is no way to know how much weight the P factor should carry; therefore, it is simply 1 for this analysis.

2.3 R and K Factors:

The R and K factors are special in this analysis as they are not spatially derived. These two factors were based on averages, the K factor for example is an average of K values for the three most common soil types in the area, provided by the RUSLE handbook (Renard et al., 1997). The R factor was also an average of published values for the Central Coast. These two remain static throughout the analysis, the R value is $250\text{MJ} \cdot \text{mm} / \text{ha} \cdot \text{h} \cdot \text{yr}$ and the K factor is 0.28.

2.4 LS Factor:

The LS factor is derived from a Digital Elevation Program (DEP). This data came from USGS 3DEP via Web Coverage Service and is natively in 30m resolution. The data, due to download limitations, was downloaded in four tiles along the study area which then were mosaiced together. Given that 30m did not provide ideal accuracy and the other spatial data was in 10m resolution, the USGS data was resampled to 10m using bilinear interpolation. The mosaic left some artifacts which were throwing off data along the boundaries of the tiles, so a uniform smoothing filter was applied to reduce this. Slope (in degrees) was then derived from the DEM using the GDAL slope algorithm with edge pixel computation. With both length and steepness available LS could be calculated using the Moore and Birch formula: $LS = (L / 22.13)^{0.4} \times (\sin \theta / 0.0896)^{1.3}$, L was the cell size and thus set to 10m.

2.5 C Factor:

Three Sentinel-2 Level 2-A tiles were downloaded for September 17th 2018 (pre Paul's Slide) covering the entire area of study. This data is in 10m resolution and the bands 4 and 8, red and near-infrared respectively, were extracted and mosaiced together. Normalized Difference Vegetation Index (NDVI) was calculated from the Sentinel-2 bands using: $NDVI = (B08 - B04) / (B08 + B04)$. C factor was then calculated in accordance with Abdelsamie et al., 2022 using the exponential decay relationship: $C = \exp(-2.5 \times NDVI)$. As was outlined in the article, values were constrained to 0.05 – 1 where 1 is bare soil and 0.05 is a dense forest.

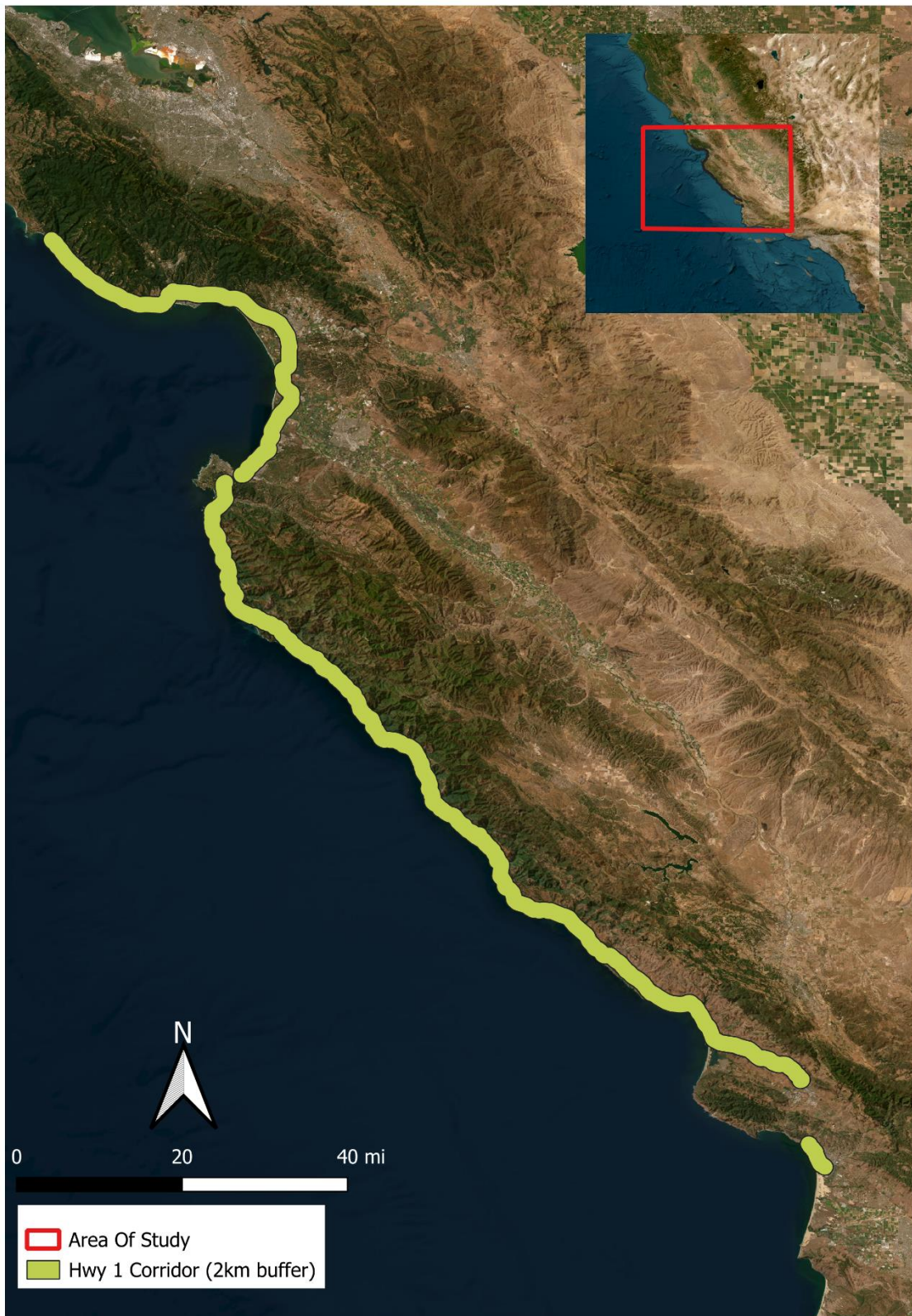


Figure 1: Area of Study

2.6 Computation and Classification:

The raster data for the C and LS factors were checked to confirm they shared the same coordinate system and resolution. They then were aligned to a common grid extent, given the C factor raster covered a larger area than the LS factor it was resampled to the same grid extent as the LS using bilinear interpolation. A no data mask was also applied during this time so that areas such as the ocean or areas with no reflectance would not deflate averages in the final analysis.

RUSLE was then computed for each cell as a cell-by-cell raster multiplication. Errors came about during this as QGIS was having trouble processing the task of multiplying the factors together at each of roughly 350 million cells. The app would frequently timeout leading to incomplete or no data in the output, so a shift was made to executing the code in a standalone python window using NumPy and GDAL libraries. The output of the script was written as a GeoTIFF file. The script will be made available in the Appendix section.

With the RUSLE raster finished it was then brought into QGIS for analysis. The raster, covering the extent of the LS factor was clipped to the 2km Highway 1 buffer to finalize the search area. The raster was then reclassified into 6 classes, Very Low <5, Low 5-20, Moderate 20-50, High 50-100, Very High 100-200, and Extreme >200. These classes were based on the Food and Agriculture Organization (FAO 1979) soil loss classification scheme. From this point any QGIS mapping was symbolizing to personal preference and for ease of the viewer.

3. Results:

3.1 Spatial Distribution of Erosion:

Figure 2, pictured below on page 7, shows the entirety of the study area along with the erosion risk raster overlayed on top. Low to moderate risk areas are roughly where one would imagine; There are large clusters of these zones around cities, Monterey and Santa Cruz most notably. These are of course going to have minimal erosion as they are graded to have a nearly unnoticeable slope. Moderate to high values can be found in the Southern portion of the search area in the more mildly sloped San Simeon as well as to the Northwest coming out of Santa Cruz. There are extreme values scattered throughout the map however by far the area with the highest density of extreme values is the section of road between Carmel and San Simeon. This comes to little surprise as this is the area which prompted this project, there are miles of Big Sur which have practically no values besides extreme or very high due to the steep terrain and at times minimal plant cover. Overall, high values are centered in areas that A, have steep slopes, and B, have minimal plant cover. That said, there is an area just North of San Simeon which I believe has been overestimated for risk, this will be touched on in the Discussion section.

3.2 Risk Class Distribution

The risk classes were split relatively evenly between low to very high risk classes sitting around 15% each, however, the extreme class had by far the largest number of cells, making up over 30%. The mean erosion rate across the study area was 168 tons per acre per year with the highest values reaching up to 900 tons per acre per year in a couple of cells. A table depicting exact results and distribution is on Page 8, figure 3.



Figure 2: Erosion Risk Along Highway 1

Risk Class	Soil Loss (t ha ⁻¹ yr ⁻¹)	Area (ha)	Area (km ²)	% of Corridor
Very Low	< 5	1,874.1	18.7	4.8%
Low	5 – 20	7,075.0	70.8	18.3%
Moderate	20 – 50	6,199.9	62.0	16.0%
High	50 – 100	5,387.2	53.9	13.9%
Very High	100 – 200	6,272.9	62.7	16.2%
Extreme	> 200	11,877.2	118.8	30.7%
TOTAL		38,686.3	386.9	100.0%

Figure 3: Risk Class Distribution Table

4. Discussion:

Upon reading the data the first time around, a mean erosion rate of 168.6 tons per hectare per year felt high, however, given the topography and nature of Big Sur it actually is more accurate than I had thought. The mean erosion rate for just Carmel to San Simeon was 285.6 tons per hectare per year, this naturally skewed the results more towards the very high class overall. While still skeptical of this data, I consulted two faculty in the Cal Poly Soil Science Department, Cody Cameron and Dr. Chip Appel. While talking with them and presenting this data, they both seemed to agree that the figure was more than likely accurate. RUSLE takes into account erosion from both sheet and rill erosion as well as larger landslides which naturally makes it produce a high figure. To further corroborate the data, I found an article published by USGS about soil erosion rates in Big Sur (Hapke, C.J. and Green, K.R., 2005). This article cites 80 thousand cubic yards per mile per year, this

when converted to tons per acre per year is shockingly close at 247 tons per hectare per year. The study area for this figure comes exclusively from Big Sur, where the highest values are found in my analysis, with only 40 or so tons per acre per year difference from the mean erosion rate I found.

The USGS report was promising for the soundness of the analysis I had performed but it was also important to check if it was predicting major landslides; this was the reason I used Sentinel-2 data from 2018 rather than the present. The main slide I was interested in predicting was Paul's Slide given its effect on the road's status for the three years it was closed. A map depicting Paul's Slide overlayed on top of the erosion risk raster can be found on page 11 in Figure 4. The erosion risk analysis seems to predict the area that would begin sliding nearly five years after the imagery was taken with a good amount of accuracy. There is a small area of the slide that was not predicted in the northwest corner of the slide and some areas were overpredicted, particularly to the southeast. With that said, many of the areas are seeing erosion currently just not to the scale of Paul's Slide and should still be considered partially successful.

Despite the successful results of this analysis, there were certainly improvements to be made. The main points to address are the two static factors in the RUSLE analysis, R and K. I struggled to find any spatial data for rainfall erosivity and therefore had to assume a static figure for the entire region based on estimations from studies in the area. Realistically this number would fluctuate heavily depending on seasons and location on the coast which could dramatically affect how certain areas were classified. Similarly, I could not find spatial data for soil erodibility, the closest I could attain was a polygon layer of all soil types in the area. It would have theoretically been possible to calculate the K factor for all soil types in the area and use that to decide a given cell's K factor however given the 10-week nature of the course, this was out of my scope. For those doing analysis using this method in the future, I would recommend at least adding spatial data for the K factor and if possible to find, the R factor would also be beneficial to have spatial data for.

Another of my main concerns with this analysis comes from an area of suspiciously high erosion risk north of and coming into San Simeon. This area, while not flat, is not nearly as steep as the Big Sur region, it is however, especially in the dry months of the year, almost completely lacking any green. The imagery I pulled from Sentinel-2 was from September, this was to avoid cloud cover interfering with reflectance from the satellite, that said, September is one of the driest months of the year. Almost the entire area, pictured on page 12 in Figure 5, is between high risk and extreme risk. This leads me to believe that because of the dead annual grass that is present there the C factor is making the equation over predict this section's risk classes. For future research I would recommend using both wet and dry season imagery to see if this makes a difference.

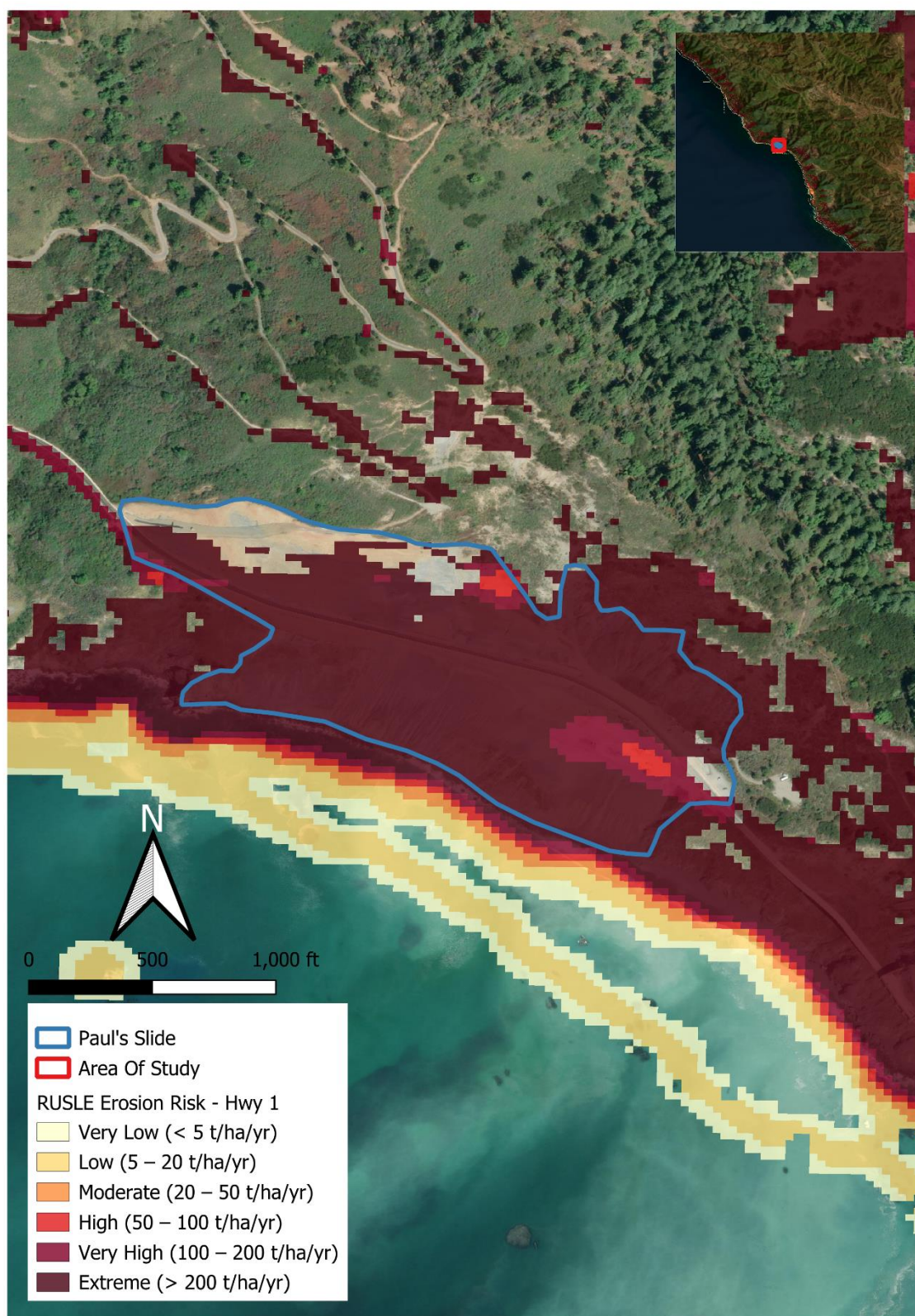


Figure 4: Pauls Slide Over Erosion Risk Raster

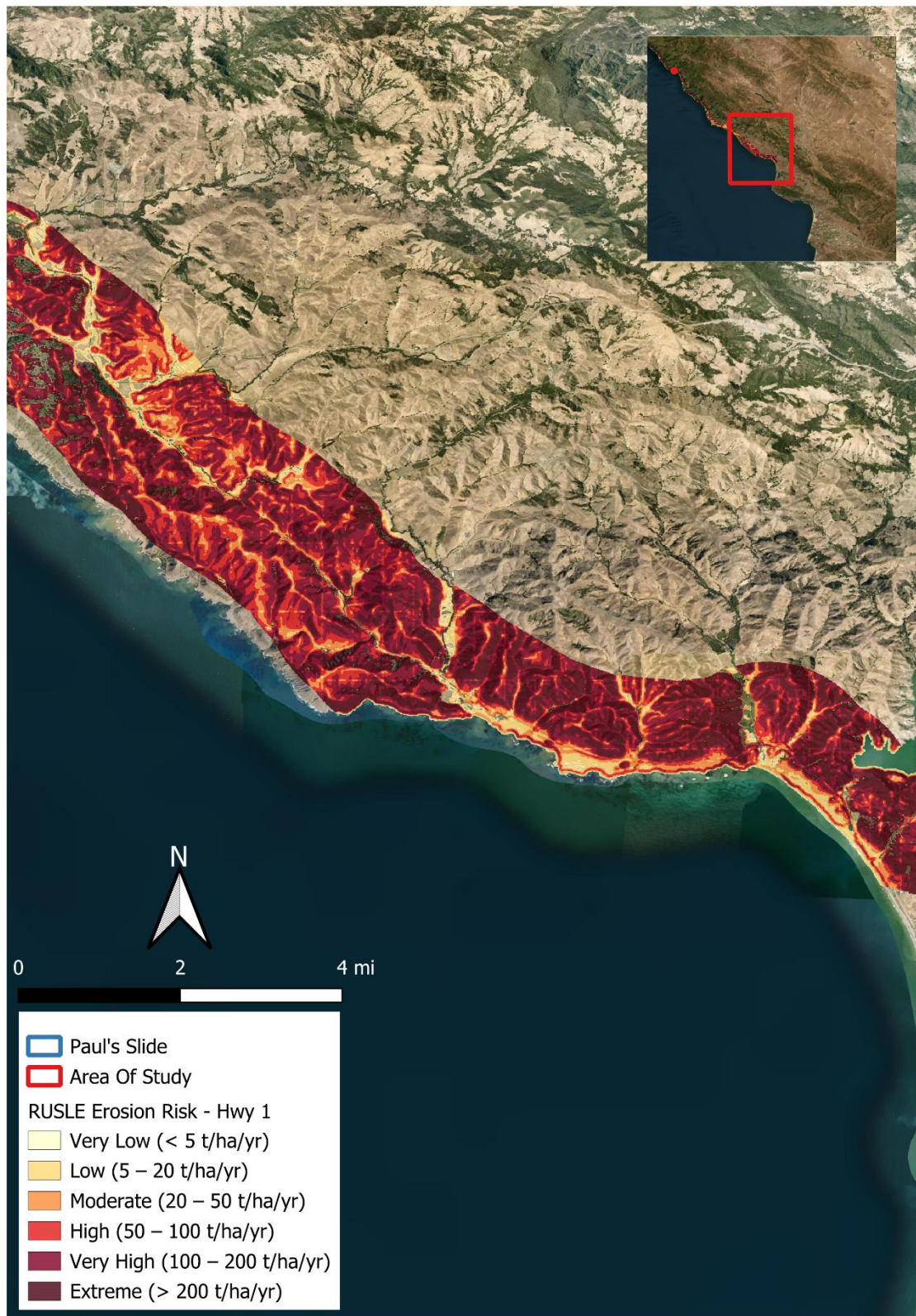


Figure 5: Potentially Overpredicted Area

5. Conclusion:

This project has yielded some potential very useful data however, I am not fully confident this is an accurate representation of erosion for the full area. The accuracy seems similar to that of USGS and has accurately predicted a handful of landslides however there are many extreme risk areas that have minimal erosion. It will be important to keep watch over these areas as years progress it is not unlikely Big Sur will have another large slide in the near future without extensive intervention from state and local government. With the handful of improvements I laid out in the Discussion section I do believe this could be a very accurate model for predicting erosion risk. Communities like San Simeon and Carmel rely very heavily on traffic and tourism that is brought in by Highway 1 so the more closures can be prevented the better, the main limitation is knowledge of where landslides may happen. With further development of this analysis workflow, I am confident that limitation can be alleviated.

6. References:

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7. Appendix:

This is the link to a GitHub page containing the Python script, annotated and edited by Claud, it outlines what data will be needed before running the script and breaks each section down in steps for clarity:

<https://github.com/MChopsticks/CentralCoastErosionRiskScript>

This is the link to a related ESRI Story Map covering a broad overview of the project:

<https://arcg.is/Cub4r1>